

RTL Design of Finite State Machine (FSM) Based Traffic Light Controller Using Verilog HDL

VLSI Design Internship – Task 4

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Internship Domain: VLSI Design

Project Title: RTL Design of Traffic Light Controller Using Finite State Machine (FSM) in Verilog HDL

Software Used: Xilinx Vivado

Language: Verilog HDL

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Acknowledgement

I would like to express my sincere gratitude to **Maincrafts** for providing me with the opportunity to complete this VLSI Design Internship. This project enhanced my understanding of digital system design, finite state machines, Verilog HDL, and FPGA design flow using Xilinx Vivado. I also thank my mentors for their guidance throughout the internship.

Abstract

A Traffic Light Controller is one of the most common applications of a Finite State Machine (FSM). This project implements a Moore FSM using Verilog HDL to control traffic signals. The controller transitions through RED, GREEN, and YELLOW states based on clock cycles. The design was simulated and synthesized using Xilinx Vivado, and the results verified the correct functionality of the FSM.

Introduction

Digital systems are widely used in modern electronic devices. A Finite State Machine (FSM) is a sequential logic circuit whose output depends on the present state and inputs. FSMs are commonly used in communication systems, processors, vending machines, elevators, and traffic control systems.

This project develops a Traffic Light Controller using a Moore FSM in Verilog HDL and verifies its functionality using Xilinx Vivado.

Objectives

- Understand the concept of FSM.
- Design a Moore FSM using Verilog HDL.
- Implement a Traffic Light Controller.
- Develop a Verilog testbench.
- Simulate the design in Vivado.
- Perform RTL synthesis.
- Verify state transitions.

Software Requirements

Software	Purpose
Xilinx Vivado	Design and Simulation

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Software	Purpose
Verilog HDL	RTL Coding
Windows 10/11 Operating System	

Theory

A Traffic Light Controller manages traffic flow by switching between different traffic signal states. The controller uses a sequential logic design where the next state depends on the current state and the clock.

The three states are:

- RED
- GREEN
- YELLOW

Finite State Machine (FSM)

A Finite State Machine consists of:

- State Register
- Next State Logic
- Output Logic
- Clock
- Reset

FSMs are classified into:

- Moore Machine
- Mealy Machin

Moore vs Mealy Machine

Moore	Mealy
Output depends only on state	Output depends on state and input
Stable outputs	Faster response

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Moore

Easier to design

Mealy

More complex

This project uses a **Moore FSM**.

Traffic Light Controller

The controller cycles through:

RED → GREEN → YELLOW → RED

Only one light is active at a time.

Design Methodology

1. Design the FSM.
2. Define states.
3. Write RTL code.
4. Create testbench.
5. Run simulation.
6. Analyze waveform.
7. Run synthesis.
8. Verify RTL.

Verilog RTL Design

The RTL module includes:

- Clock input
- Reset input
- State register
- Counter
- Output logic

The design uses:

- Sequential always block
- Combinational always block

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- Case statement

Testbench

The testbench performs:

- Clock generation
- Reset generation
- Simulation execution
- Functional verification

Simulation Results

The simulation verifies that:

- Reset initializes the FSM.
- RED state is active first.
- The controller transitions to GREEN.
- GREEN transitions to YELLOW.
- YELLOW returns to RED.
- The cycle repeats continuously.

Waveform Analysis

The simulation waveform shows:

State	Binary	Decimal
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RED	100	4
-----	-----	---

GREEN	001	1
-------	-----	---

YELLOW	010	2
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Sequence observed:

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RED

↓

GREEN

↓

YELLOW

↓

RED

↓

GREEN

↓

YELLOW

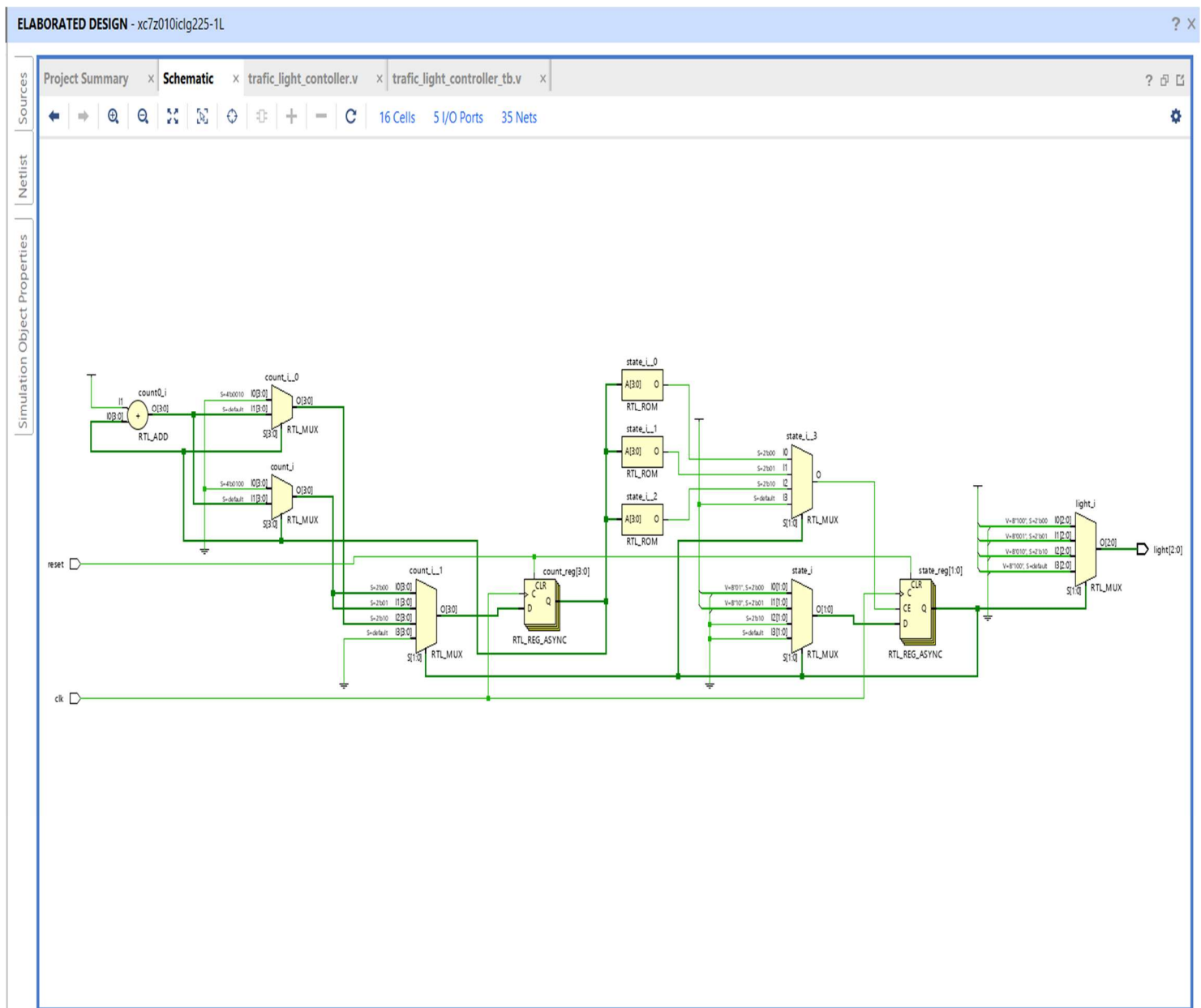
The waveform confirms that only one traffic light is active at any given time.

RTL Schematic

After RTL elaboration in Vivado:

- Inputs: clk, reset
- Outputs: light[2:0]
- Internal logic:
 - State register
 - Counter
 - Output decoder

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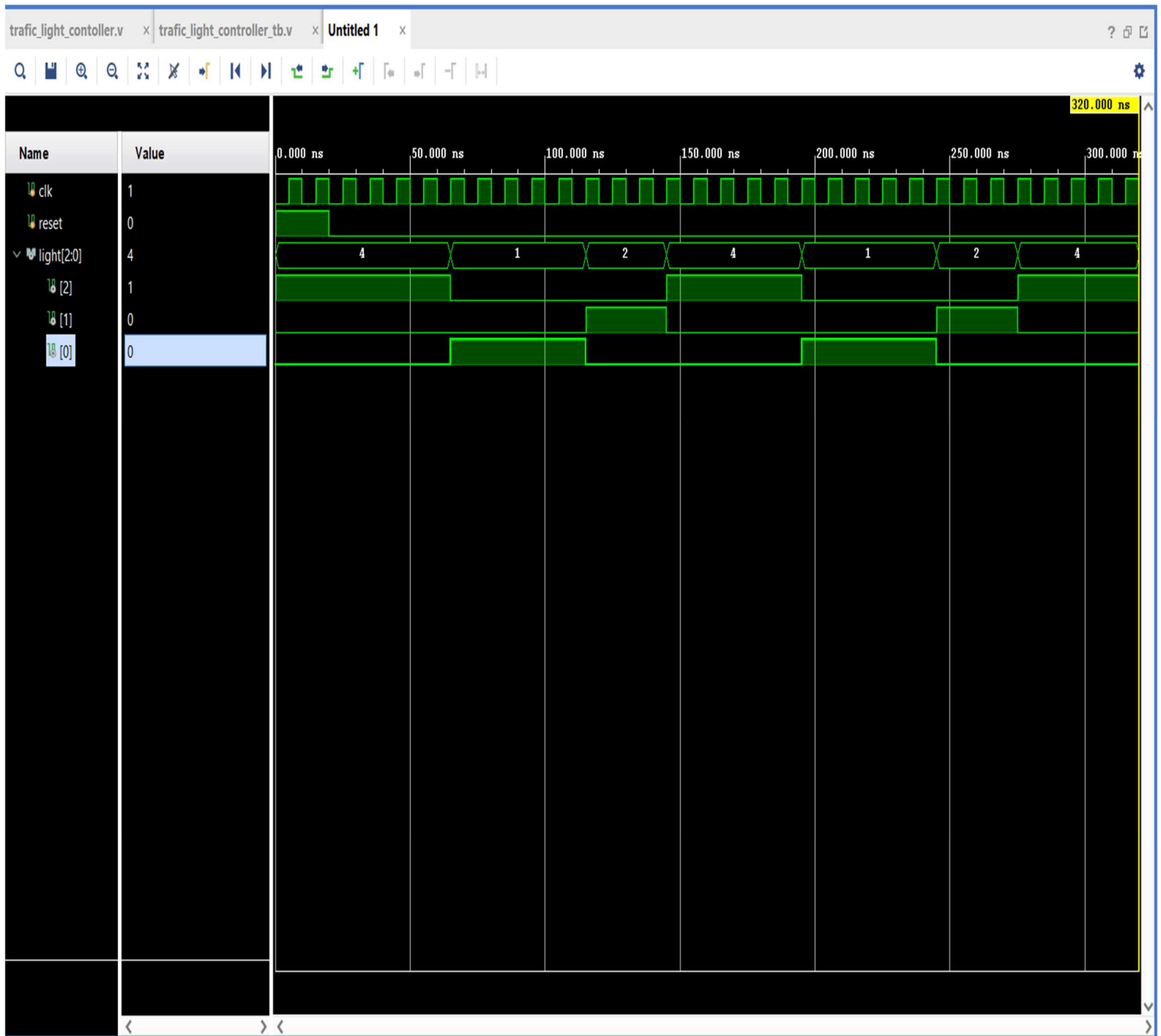
Synthesis Results

The design synthesized successfully without errors.

Key observations:

- RTL generated correctly.
- State transitions implemented.
- Resource utilization is minimal.

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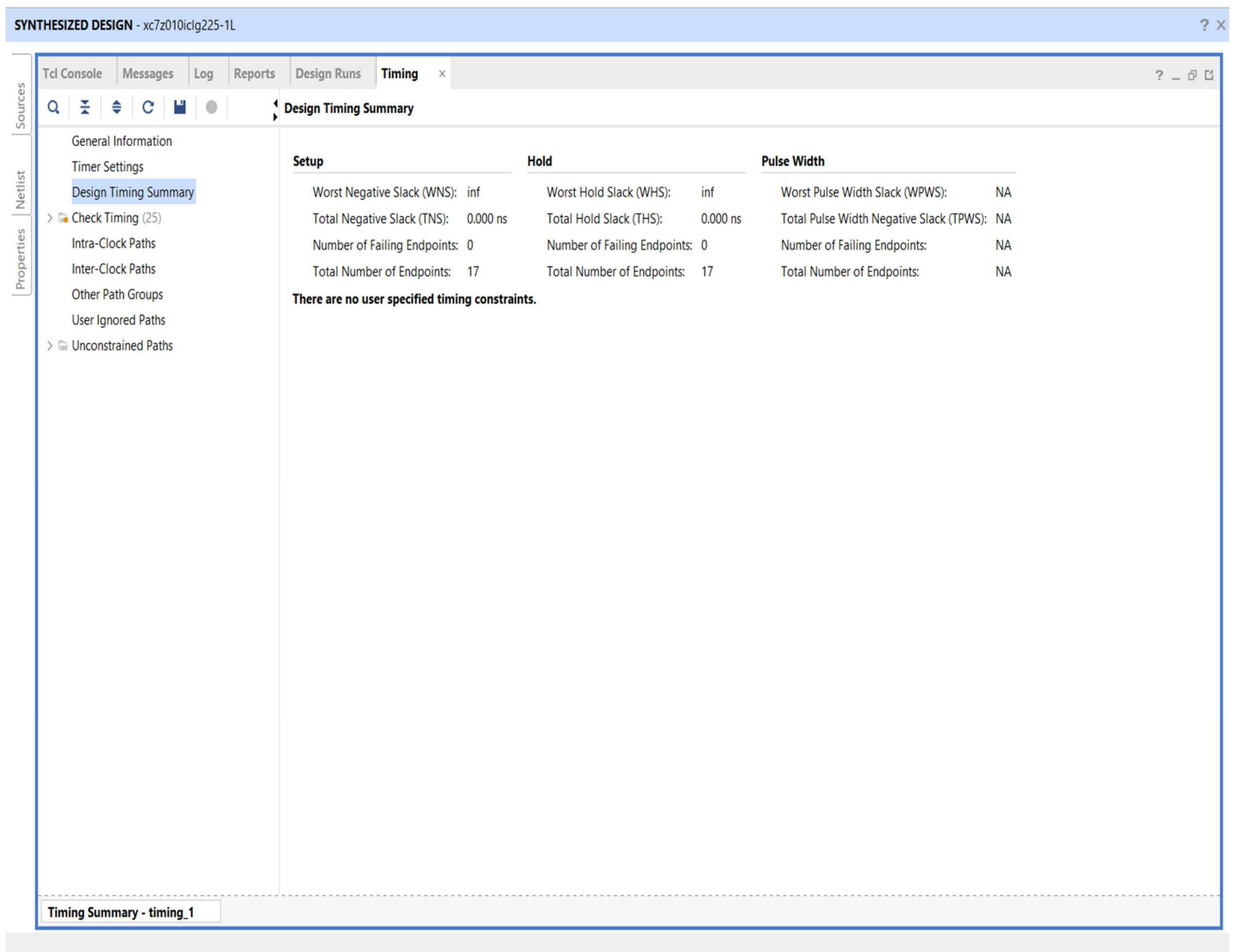
Timing Analysis

Timing Summary:

- $WNS > 0$
- $TNS = 0$

The design meets timing requirements.

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Applications

- Smart traffic control
- Railway signaling
- Industrial automation
- Embedded systems
- Digital controllers
- FPGA-based control systems

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Advantages

- Simple FSM implementation
- Low hardware utilization
- Reliable sequential operation
- Easy to modify
- Suitable for FPGA implementation

Future Scope

- Add pedestrian crossing signals.
- Control multiple road intersections.
- Implement sensor-based adaptive timing.
- Interface with FPGA development boards.
- Add emergency vehicle priority.

Conclusion

The Traffic Light Controller was successfully designed using a Moore Finite State Machine in Verilog HDL. Behavioral simulation in Xilinx Vivado verified the correct sequence of traffic light transitions, and synthesis confirmed that the design was hardware implementable. This project strengthened practical skills in RTL design, FSM implementation, Verilog HDL coding, simulation, and digital system verification.

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